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# Experiment 8

## Nmap To Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP)

**Aim:-** To discover live host discovery using nmap by performing ARP, ICMP, TCP, UDP using host discovery scan on Anyhacker platform

**Algorithm:-**

Step ①: Start Anyhacker machine and identify target IP address / range.

Step ②: Open terminal

Step ③: Perform ARP scan to discover host on local network

Step ④: Perform ICMP echo scan

Step ⑤: Perform a TCP SYN scan to discuss hosts responding to TCP

Step ⑥: Perform UDP scan to identify UDP Port

Step ⑦: Observe and record IP address of live host

Step ⑧: Compare the result obtained

Step ⑨: Stop Anyhacker.

**Program:-**

① ARP host discovery

sudo nmap -PR <target-ip/range>

[-PR performs ARP ping scan]

③ ICMP Host discovery

Sudo nmap -PE <target-ip/range>

[-PE - uses ICMP echo requests]

④ TCP SYN Host discovery

Sudo nmap -PS80, 443 <target-ip/range>

[-PS - option for TCP SN discovery on specific ports]

⑤ UDP Hosts discovery

Sudo nmap -PU53, 161 <target-ip/range>

⑥ Analyze results

* Nmap displays the hosts that are detected as living up
* Target IP address
* Host status
* Response / latency information
* Scanned hosts.

Result:

Thus, NMAP was successfully used to discover to live hosts on target network using ARP ICMP, TCP, SYN and UDP - host - discovery techniques.